The U-Dimension Theory: A Speculative Model of Multiple Universes

Abstract

The U-Dimension Theory proposes that reality contains an additional dimension beyond the familiar spatial dimensions and time. Instead of describing an event with only spatial coordinates and time, every object also possesses a **universe coordinate**, denoted by **u**. Each whole-number value of **u** represents a complete universe with its own independent space and time. Between these universes lie voids possessing unique physical properties unlike anything found in ordinary spacetime.

This theory is speculative and intended as a conceptual framework for exploring higher-dimensional physics.

The Dimensions of Reality

In this model, reality consists of six coordinates:

- **x** – horizontal position
- **y** – forward/backward position
- **z** – vertical position
- **d** – density (a physical property associated with matter)
- **t** – time
- **u** – universe coordinate

Unlike the first three spatial dimensions, the **u-dimension** does not describe movement within a universe. Instead, it identifies which universe an object occupies.

An object's location may therefore be represented as:

$$(x,\ y,\ z,\ t,\ u)$$

where **u = 0** corresponds to our universe, **u = 1** to another universe, and so on.

The Structure of the U-Dimension

Only whole-number values of **u** represent complete universes.

Examples include:

- u = 0
- u = 1
- u = 2

- ...

The regions between these integers are known as **voids**.

For example:

- u = 0.37
- u = 1.82
- u = 7.14

are not universes but void regions.

The universes themselves cannot collide because they are effectively infinitely separated within the u-dimension.

Properties of the Void

The void is unlike ordinary space.

Its defining property is that its effective three-dimensional volume is zero.

When matter enters the void, it cannot maintain its ordinary size. Instead, every spatial dimension contracts to an extremely small positive value approaching zero.

Rather than becoming exactly zero-dimensional, an object shrinks until its dimensions are almost point-like.

Time similarly contracts toward an extremely small value, making travel through the void appear nearly instantaneous to the traveler.

Objects may remain inside the void if prevented from reaching another universe, effectively becoming trapped between universes.

Motion Between Universes

Matter may move from one universe to another under special circumstances.

Artificial technology may intentionally transfer objects through the void, while naturally occurring events such as extremely unstable black holes may provide access only to the void itself rather than directly to another universe.

Movement between universes conserves matter.

If a 10 kg rock leaves Universe 0 and enters Universe 1:

- Universe 0 loses 10 kg.
- Universe 1 gains 10 kg.

No duplicate objects are created.

Independent Time

Each universe possesses its own independent timeline.

For example, a traveler may leave Universe 0 when its local time equals 10, spend time in Universe 1 while its own clock advances, and later return to Universe 0 to find that its local time has not advanced during the traveler's absence.

Thus, time is local to each universe rather than shared across all universes.

Divergence of Universes

Every universe begins as an identical copy immediately following its own Big Bang.

Initially, every planet, star, and object is identical.

As time passes, tiny differences accumulate.

Artificial transfers of matter between universes further increase these differences.

For example, moving Earth's Moon from Universe 0 into Universe 1 would leave Universe 0 without a Moon while Universe 1 would contain two Moons, causing the histories of both universes to diverge permanently.

Gravity

Within ordinary universes, gravity behaves normally and depends on mass, allowing everyday observations to remain consistent.

The void, however, places matter into an extremely compressed state due to its zero-volume environment. This compression may produce unusual gravitational effects unique to the void itself without altering ordinary gravitational behavior inside complete universes.

Infinite Universes

The theory proposes that the process responsible for the Big Bang is not unique.

Instead, infinitely many Big Bangs may occur, each producing a new universe located at another whole-number position along the u-dimension.

Consequently, the sequence

$u = 0, 1, 2, 3, \ldots$

extends without limit, producing an infinite collection of independent universes.

Conclusion

The U-Dimension Theory presents a speculative picture of reality in which universes are arranged along a higher-dimensional coordinate. Between them lies a zero-volume void capable of compressing matter to an almost point-like state while allowing transfers between universes under extraordinary circumstances.

Although this model differs significantly from accepted modern physics and has not been experimentally verified, it provides an internally structured framework for exploring concepts such as parallel universes, higher dimensions, independent timelines, conservation of matter across universes, and the role of extreme environments such as black holes.

Whether correct or not, the theory illustrates an important aspect of scientific thinking: beginning with a set of rules, following their logical consequences, and refining the model through questions and consistency.